Emma E Bonham

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LinkedIn: www.linkedin.com/in/BonhamEmma

Google Scholar: https://scholar.google.co.nz/citations?user=NzoFujMAAAAJ&hl=en

Technical Skills

MATLAB | R Studio & R Shiny | ArcGIS Pro | Qualtrics Surveying | Data & Statistical Analysis
Microsoft & Google Suite | Field & Lab Work | Project Management | Technical Writing & Presentation

Education

Ph.D. Civil, Environmental, and Sustainable Engineering **Graduation: December 2025**

Assessing Abandoned Mine Lands through Water Quality Analysis, Predictive Modeling, and a Vulnerability Mapping Tool *– Ph.D. Dissertation*

Ira. A. Fulton Schools of Engineering, Arizona State University GPA: 4.00

MS Civil, Environmental, and Sustainable Engineering **Graduation: May 2022**

Assessing Organic Farm Nutrient Management *– Master's Thesis*

Ira. A. Fulton Schools of Engineering, Arizona State University GPA: 3.75

BSE Environmental Engineering, Sustainability Minor **Graduation: May 2021**

Assessing Urban Agricultural Practices in Desert Cities *– Barrett Honors Thesis*

Ira. A. Fulton Schools of Engineering, Barrett the Honors College at Arizona State University GPA: 3.72

Relevant Coursework *(Official transcript available upon request)*

GIS Methods | Probability & Statistics for Engineers | GIS and Public Health

Environmental Data & Analysis | Data Programming in R | Data Wrangling | GIS for Hazards Analysis

Research and Experience

Hamilton Lab Group | Arizona State University – *Graduate Research Associate* **Aug 2023 – *Present***

- Analyzing environmental data to predict the locations of undocumented abandoned mine lands and creating a GIS-integrated R Shiny application for prioritizing hazard verification and remediation activities.
- Overseeing the management of a collaborative project involving three universities.
- Developing and delivering comprehensive, actionable research presentation and reports to stakeholders on behalf of a tri-university collaboration

Muenich Lab Group | Arizona State University – *Graduate Research Assistant* **Sep 2019 – Jul 2023**

- Designed and conducted an organic farm sustainability study, surveying over 100 farm operators across the United States, and managed soil sample collection and return.
- Contributed to an urban stormwater study by performing fieldwork and laboratory analyses, including the collection of water samples and execution of laboratory tests.
- Transcribed telephone interviews for a peer's PhD dissertation, gathering information from farmers regarding their management practices and perceptions of operational outcomes.

United States Department of Agriculture – *Research Intern* **May 2020 – Aug 2022**

- Directed a research project evaluating the sustainability of nutrient management practices on urban farms in Arizona.
- Designed and implemented surveys to evaluate water use efficiency in landscaped areas and residential gardens.
- Coordinated scientific outreach initiatives by overseeing academic social media platforms and organizing webinars featuring agricultural specialists to advance agricultural research and educational engagement.

Publications

Zanin, A. C., Tietsort, C. J., Muenich, R. L., & **Bonham, E. E.** (2023). Irrational rationality in organizational decision-making: sustainability discourses and material constraints of U.S. urban desert farmers. *Journal of Applied Communication Research*, *52*(1), 129–149. https://doi.org/10.1080/00909882.2023.2220779

Bonham, E. E., Muenich R. L., Xu T., Hamilton K. A. (***Publication in review***). *Water Quality Signatures for Prioritizing Abandoned Mine Land Ground-truthing in the Southwestern United States*.

Bonham, E. E., Muenich R. L., Xu T., Hamilton K. A. (***Preparing for publication***). *Predicting Mining Activity Using Water Quality and Environmental Data in the Southwestern United States*.

Bonham, E. E., Sutherland V., Kollus K. M., Grajal-Puche A., Alsina-Harris M., Bos Orent E. H., Driver E. M., Muenich R. L., Folch D. C., Propper C. R., Conroy-Ben O., Halden R. U., Hamilton K. A. (***Preparing for publication***). *Vulnerability Scoring System to Prioritize and Manage Land Remediation with a Focus on Abandoned Mines in Arizona.*

Bonham, E. E., Williams C., Zanin A. C., Muenich R. L. (***Preparing for publication***). *Assessing Organic Farm Nutrient Management.*

Presentations

Bonham E.E., Sutherland V., Kollus K. M., Grajal-Puche A., Alsina-Harris M., Bos Orent E. H., Driver E. M., Muenich R. L., Folch D. C., Propper C. R., Conroy-Ben O., Halden R. U., Hamilton K. A. (2025, December 7-11). *A Multi-Criteria Vulnerability Mapping Tool (VMT) for Prioritizing Abandoned Mine Land Remediation Integrated with Predictive Modeling* [Oral presentation]. Society for Risk Analysis, Washington D.C., United States.

Bonham E. E., Sutherland V., Kollus K. M., Grajal-Puche A., Alsina-Harris M., Bos Orent E. H., Driver E. M., Muenich R. L., Folch D. C., Propper C. R., Conroy-Ben O., Halden R. U., Hamilton K. A. (2025, October 5-8). *A Vulnerability Scoring Tool for Prioritizing Abandoned Mine Land Remediation in Arizona* [Oral presentation]. National Association of Abandoned Mine Lands, Gulf Shores, AL, United States.

Bonham E. E., Sutherland V., Kollus K. M., Grajal-Puche A., Alsina-Harris M., Driver E. M., Muenich R. L., Folch D. C., Propper C. R., Conroy-Ben O., Halden R. U., Hamilton K. A. (2024, December 7-12). *Developing a risk-scoring method to prioritize abandoned mine land remediation in Arizona* [Poster presentation]. Society for Risk Analysis, Austin, TX, United States.

Bonham E. E., Muenich R. L. (2022, November 30-December 1). *Assessing Organic Farm Nutrient Management* [Poster presentation]. Organic Growers Summit, Monterey, CA, United States.

Bonham E. E., Muenich R. L. (2022, November 16-17). *Assessing Organic Farm Nutrient Management* [Poster presentation]. Sustainable Agriculture Summit, Glendale, AZ, United States.

Bonham E. E., Muenich R. L. (2022, July 21-24). *Assessing Organic Farm Nutrient Management* [Poster presentation]. Sustainable Agriculture Education Association, Columbus, OH, United States.

Bonham E. E., Muenich R. L. (2022, April 22). *Assessing Organic Farm Nutrient Management* [Poster presentation]. Master's Opportunity for Research in Engineering, Tempe, AZ, United States.

Bonham E. E., Muenich R. L. (2020, November 20). *Assessing Urban Agricultural Practices in the Desert Cities* [Poster presentation]. Fulton Forge Student Research Expo, Tempe, AZ, United States.

Certifications

Engineer-in-Training (EIT) Certification	**Jun 2022**

Memberships and Extracurriculars

Society for Risk Analysis	**Dec 2024**
Tau Beta Pi Engineering Honor Society	**Apr 2024**
American Society of Civil Engineers (ASCE)	**Aug 2019**
Society of Water and Environmental Leaders (SWEL)	**Aug 2019**
Society of Women Engineers (SWE)	**Aug 2017**